

Capstone Project Proposal

Title and Team Metadata

Project Title: GM Simulator

Project Size: Solo

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Stakeholders: Bradley Allen, Addison Gage, Sara Alhusaynat

Studio Session: 1

Problem Framing and Research Question

Background

The history of baseball spans over a century, demonstrating continuous changes from its early twentieth-century origins to the modern era. Historically, managerial methodologies relied heavily on intuitive "gut feelings" and traditional observations; however, the contemporary landscape has transitioned toward a more analytical framework. A pivotal moment in this evolution was the "Moneyball" phenomenon, popularized by the Oakland Athletics. This strategic paradigm shift prioritized quantitative metrics over subjective human assessment, utilizing calculated data points to optimize roster construction and lineup efficiency. This project is driven by a deep-seated interest in the underlying mathematical architecture of the sport, seeking to determine the efficacy of these data-driven strategies in achieving competitive advantages.

Vision

The overarching vision of this project is to equip baseball organizations with a predictive analytical environment, effectively translating raw environmental variables and individual player metrics into actionable intelligence to optimize strategic roster management and field positioning.

Mission

The primary mission of this project is to build a comprehensive data pipeline and data storage infrastructure, to create a simulation engine. By synthesizing official MLB statistical

datasets with climate forecasts and unique stadium spatial parameters, the initiative aims to provide probabilistic game outcome distributions. Through a dynamic web based dashboard designed for general managers, offering comparative simulation capabilities between opposing teams.

Research Question

Main Question

To what degree is it possible to forecast the result of a baseball game by analyzing individual player performance history, environmental factors, and collective team health to estimate a team probability of victory?

Other Questions

1. Is it feasible to forecast individual plate appearance results by synthesizing historical performance data with specific environmental variables?
2. How significantly do external environmental factors contribute to the final outcome of a matchup?
3. In the context of roster management, what relative weight should be assigned to a player's current performance trends versus their career statistics against a specific pitcher?
4. Given a particular pitching opponent, what is the mathematically ideal batting order for a lineup?

Adjustment

The initial charter submitted prior to this proposal focused on a different subject. However, extensive preliminary research revealed a significant lack of available data, which would have compromised the integrity of the research question and rendered the project unachievable. Consequently, the focus has shifted to a secondary idea that was previously proposed.

Impact and Stakeholders

Impact

If successful, this project demonstrates a blueprint for contextual sports analytics. Organizations benefit from a sports-analytics framework capable of running "what-if"

environmental simulations allowing front offices to optimize starting lineups, pitching rotations, and in-game strategic shifts based on real-time weather changes and park dimensions.

Stakeholder

Key stakeholders for this project include baseball general managers and front-office personnel, who can leverage these analytical insights to inform data-driven decisions concerning starting lineups and defensive positioning. Additionally, classmate stakeholders Bradley Allen, Addison Gage, and Sara Alhusaynat will provide guidance and information to ensure the project remains on track and aligned with its core objectives.

Shared Theme

Prior to the shift in focus, the shared project theme centered on personal health and medical care. However, further research of the previous topic showed that the proposal was not achievable, so there was a change in the topic. Within this new area of expertise, I look forward to receiving guidance and feedback from my classmates. Despite the departure from the previous theme, I am hopeful that their insights will help identify diverse, achievable routes and ideas for this project within the remaining timeframe.

Data Sources and Access Plan

Data Sources				
Name	Source	Data	Volume	Approval
Pybaseball	pybaseball	Python scraper that uses Statcast, Baseball Reference, and Fan Graphs to important	Over 180,000 individual plate-appearance events per MLB season, culminating in a processed dataset of nearly	Needed for Professor but able to access

			320,000 distinct at-bat event rows. Data extraction happens via batch scripts.	
MLB-StatsAPI	https://github.com/toddrob99/MLB-StatsAPI	This API scraper collects information from the live MLB website from live game scores to player outcomes	It is expected to have 2430 games per year for a 2 season span. We expect to have around 5,000 rows of games.	Needed for Professor but able to access
Open-Meteo API	https://open-meteo.com/en/docs/historical-weather-api	Has information about all weather and other weather facts around the country.	4,860 unique rows linked directly 1:1 with the historical game log fact table.	Needed for Professor but able to access
MLB Ballpark	https://www.kaggle.com/datasets/paulrjohnson/mlb-ballparks	This Kaggle dataset provides comprehensive stadium specifications, including detailed field dimensions.	A static matrix of 30 rows by 12 structural attributes, updated once annually.	Needed for Professor but able to access

Data Engineering Plan

Infrastructure and Tools

- Database Engine: A relational PostgreSQL instance provisioned and hosted continuously on Railway. PostgreSQL enforces strict relational keys, schema validation, and handles quick numerical join across the tables
- Database Management: Beekeeper serves as the primary visual editor and connects through the database url.

Repository Layout and Data Tiering Strategy

- data/raw: Completely immutable. The ingestion pipeline queries external APIs and saves raw data here. These files are never modified, serving as a clean local backup.
- Railway/Postgres: This facilitates the integration of relational tables and database views, ensuring streamlined data storage and maximum retrieval efficiency.

Collect of Data

To facilitate the collection of the information from both static data sources and external APIs, I will implement a dual-path collection and ingestion strategy to populate the project's data architecture.

Stagnant Files

MLB Ballpark Data: Given that this is a static dataset, it will be downloaded as a one-time acquisition. This process ensures all necessary stadium specifications are captured before being integrated into the stadium database tables.

API Data Collection

The remaining data sources will adhere to a standardized ingestion protocol for integration into the Railway-hosted storage architecture. This phase utilizes an [ingest.py](#) script, which executes scheduled cron jobs to scrape data from the MLB and Open-Meteo weather APIs. These harvested files are then persisted as immutable backups within the data/raw directory of the GitHub repository. Subsequently, the [features.py](#) script performs the requisite join operations to synthesize game and meteorological data prior to database

insertion. The final ingestion process employs scheduled Python scripts that, upon completion of data cleansing, transmit records to the PostgreSQL system. To maintain data exclusivity and prevent redundancy, the pipeline utilizes SQL INSERT statements with ON CONFLICT clauses, ensuring that each data point remains unique within the production environment.

Planned Analysis

Instead of predicting a final game score directly using historic team averages, this project models the game from the ground up by combining Machine Learning with a Stochastic State-Machine.

Analysis

- We train an XGBoost Classifier to predict the precise probability distribution of a single plate appearance (PA). Inputs: 15-game rolling player form (momentum), pitcher situational stats, outfield fence geometry, and localized climate vector math (air density + wind speed/direction components).
- We feed those dynamic probabilities into a custom baseball rules engine (src/[simulator.py](#)). The script manages innings, outs, and base-runner advancements. For every matchup, it calls the XGBoost model to get the context-specific probabilities, samples an outcome, updates the diamond, and cycles through the lineup.

Evaluation & Validation Strategy

To demonstrate that the integration of micro-environmental variables enhances predictive accuracy, the system architecture maintains a stringent validation framework:

- Rigorous Seasonal Partitioning: Model training is conducted *solely* using data from the 2024 MLB Regular Season (approximately 180,000 plate appearances), while validation is performed *exclusively* on the 2025 MLB Regular Season.
- Elimination of Target Leakage: Predictive features for any simulated validation instance are derived strictly from historical data points generated *prior* to the scheduled date of the game.
- Comparative Benchmarking: System performance is evaluated against a Naive Baseline (based on league-wide frequency averages) and a Traditional Performance Baseline,

which utilizes standard regressions of seasonal player statistics without consideration for meteorological or stadium-specific constraints.

Program Integration (Five Pillars)

Data Engineering

- Python scripts that automatically pull data from separate APIs , clean them, and sort them into localized file tiers It then pushes these clean matrices into a live, cloud-hosted PostgreSQL database on Railway managed via Beekeeper Studio.

Analytics & Machine Learning

- It uses a two-tiered predictive setup. First, an XGBoost Multi-Class Classifier calculates the exact mathematical probabilities of a single at-bat outcome based on situational data. Second, a custom Monte Carlo state-machine (src/simulator.py) runs those probabilities to map out full game-score distributions.

Statistics & Research Design

- The project enforces a strict Sequential Time-Series Split, training *only* on 2024 season data and testing *only* on 2025 data. It uses rolling historical lag functions to guarantee zero target leakage

Visualization & Communication

- Translates complex code and backend databases into actionable human to connect the Python simulation engine to an interactive, public Streamlit web application dashboard. This front-end allows a non-technical General Manager to drop down active lineups, adjust weather sliders (like wind speed or temperature), and instantly see visual probability curves of the game's outcome.

Ethics & Responsible Use

- Because predictive models directly impact player evaluations, salaries, and playing time, this project rejects "black-box" systems. The application surfaces global feature importance charts and decision pathways so executives can see exactly *why* a model projected a certain outcome, ensuring fairness and interpretability.

Ethical and Risk Notes

Ethical

Predictive tools used by front offices directly impact player compensation, arbitration values, and field time. If a prediction model operates as an opaque "black box," it can create unfair biases against certain player profiles. To ensure transparency, the dashboard surfaces global feature importance metrics and local decision paths. This allows users to see exactly how much an environmental adjustment (e.g., wind speed) influenced a player's projected performance. Furthermore, the model limits inputs to on-field metrics, remaining blind to player age, contract size, or demographic backgrounds to maintain objective, performance-based predictions.

Know Data Limitation and Collection Risk

- Undocumented API Changes: The MLB API is internally managed and can change its data format without warning, causing ingestion scripts to crash.
- API Rate Limiting & Timeouts: Pulling thousands of historical game and weather rows requires making tens of thousands of web requests, which can lead to blocked IPs or dropped connections.
- Missing Relational Rows: If a specific hourly weather timestamp or stadium coordinate fails to match during database merges, standard data joins will completely drop that entire game's history.

Timeline and Milestones

Week	Target Milestone	Active Tasks & Deliverables	Phase-Specific Risk
Week 4	M1-proposal	Commit CHARTER.md; spin up Railway PostgreSQL	Database connection timeout or hosting configuration errors.

		instance; configure Beekeeper Studio.	
Weeks 5–6	<i>Pipeline Dev</i>	Write src/ingest.py; download 2024–2025 raw data to data/raw/.	Undocumented MLB API schema changes crash the script.
Week 7	M2-data summary	Clean data rows; run missingness checks; compile formal summary report in deliverables/.	Missing hourly weather timestamps drop entire games during database joins.
Weeks 8–9	<i>ML Training</i>	Write src/features.py and src/simulator.py; build rolling stats, wind vectors, and the Monte Carlo loop.	Data leakage (\$t-1\$boundary broken) causes fake, inflated model accuracy.
Week 10	M3-poster draft	Optimize multi-class XGBoost models; draft and present the initial visual technical poster.	High simulation variance or slow loop execution times across 10,000 runs.
Week 11	<i>UI & API Dev</i>	Build production FastAPI endpoints in src/app.py; assemble the Streamlit frontend.	Streamlit interface lag when calling the backend simulator engine.

Week 12	M4-writeup-draft	Conduct extreme weather stress-testing; compile formal manuscript draft in deliverables/.	The model acts as an uninterpretable "black box" during peer PO critiques.
Week 13	Refinement	Fix front-end UI bugs; scale deployment configs; update Iteration Reviews based on feedback.	Cloud database connection pool exhaustion under continuous hosting.
Week 14	M5-final	deploy the finalized application live; complete the capstone presentation.	Production deployment crash on the live server right before evaluation.

DS3 Process and Tooling

GitHub repo URL: https://github.com/perezamp15-hue/data510-Betting_The_Right_Way

GitHub Projects (v2) board URL: <https://github.com/users/perezamp15-hue/projects/1>

Stakeholder Communication Strategy: To facilitate effective engagement with shareholders, project updates and system modifications will be disseminated via a dedicated Discord server. This platform will serve as the primary channel for soliciting technical feedback and strategic advice to inform the iterative development of the project.